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Extended Dynamics Observer for Linear Systems with Disturbance*

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Abstract

This paper aims at designing observer for single-input-single-output (SISO) linear finite-dimensional systems corrupted by general disturbances. A new observer, referred to as *extended dynamics observer* (EDO), is proposed to estimate both state and disturbance simultaneously. The only assumption for the observer design is that the plant with unknown disturbance is observable. The disturbance is treated by combination of high-gain and internal model contained in the extended dynamics. The main advantage of the developed method lies in the maximum utilization of the prior information of the disturbance. The more accurate the prior disturbance dynamics we have, the better performance the observer would be. The well-posedness of EDO is established and the theoretical results are validated by numerical simulations.

Keywords: Active disturbance rejection control, internal model principle, disturbance, observer, system dynamics.

1 Introduction

The observer design is one of the important topics in modern control theory, which has various applications in output feedback control, system monitoring, process identification and fault detection, among many others. The tolerance of external disturbance and system uncertainty becomes a major concern in observer design. Many methods for observer design have been developed for uncertain systems. These include but not limited to the sliding mode observer [15], [20], the high-gain observer [14], the unknown input observer [21],[8], the observer based approach in internal model principle [18], and the adaptive observer [12], to name just a few. If the system state and

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the disturbance were estimated effectively online, the control system with disturbance would become almost known, which makes the feedback control design much easier. In particular, when the disturbance lies in the control channel, the disturbance can be compensated by its estimation directly. The observer design is therefore very important to the controller design, in particular to the estimation/cancellation strategy in the active disturbance rejection control (ADRC) [7].

The extended state observer (ESO), as a key part of ADRC, can estimate both the disturbance and system state simultaneously by using the high-gain technique. Regarding the un-modeled dynamics of system as a disturbance, ESO is similar with the high-gain observer [6, 4, 16, 5]. Actually, ESO is always working provided the derivative of the disturbance is bounded. However, every coin has two sides. On the one hand, the model free characteristic leads to the strong robustness to the uncertainty and disturbance, and on the other hand, it may waste more or less some useful prior information that we have already known about plant and disturbance. The waste of the prior disturbance information also exists with varying degrees in other observers such as the sliding mode observer [15] and the extended high-gain observer [3], [19]. Actually, in engineering applications, we are not always completely ignorant of the disturbance. Some information like smoothness, boundedness, particularly some dynamic information of the disturbance are available sometimes. This prior information might be useful or even valuable for control purpose. A typical example is the harmonic disturbance where the known frequencies are very useful in internal model principle (IMP) which is an elegant approach to robust output regulation [9] yet are completely wasted in ADRC. However, the disturbance in IMP is almost known. Precisely, the dynamics or the dynamical structure of disturbance are required to be known in IMP, which blocks the general disturbance out the door of IMP although there are a few extensions of IMP ([11], [10]). In one word, there is still much room for improvement for both ADRC and IMP but has not been noticed and emphasized at least in existing literature from the best of our knowledge.

In this paper, we propose a new idea to the observer design which can make maximum use of the prior disturbance information. Our starting point for observer design is the sufficient utilization of the prior disturbance dynamics. This enables us to develop a fundamental principle to design observer via online measurement from the plant and prior information of the disturbance. We believe that a good observer should possess not only the strong robustness to the disturbance but also the ability to make sufficient use of all the valuable prior information of the disturbance. The more the prior information is properly used, the better performance of the observer would be. Even the prior information is insufficient, the observer can still do its best. In this spirit, a new observer, referred to as *extend dynamics observer* (EDO), is designed in this paper to estimate both disturbance and system state simultaneously. The EDO inherits almost all the advantages from the ESO like model free characteristic yet can properly utilize the prior information not only about the control plant but also the disturbance. If all the prior dynamic information about the disturbance is available, the EDO can achieve a zero steady-state error. The EDO takes advantages of the IMP and ADRC and remedies the defect of each of them.

Consider the following SISO system:

$$\begin{cases} \dot{x}(t) = Ax(t) + Bd(t), \\ y(t) = Cx(t), \end{cases} \tag{1.1}$$

where $A \in \mathbb{R}^{n\times n}$, $C \in \mathbb{R}^{1\times n}$, $B \in \mathbb{R}^n$, $y(\cdot)$ is the measurement and $d(\cdot)$ is the total disturbance. In this paper, all the unknown signals are referred to as disturbances which may contain system uncertainties and external disturbances, that is, the disturbance $d(t)$ can be decomposed into $d(t) = d_0(t) + f(x(t))$, where the $d_0(\cdot)$ is the external disturbance and $f : \mathbb{R}^n \to \mathbb{R}$ is an unknown function. Since the estimation of the state and disturbance is a key issue in the feedback linearization ([7, 3]), we focus only on the observer design in this paper for system (1.1).

The rest of the paper is organized as follows. In the next section, Section 2, we consider the observability of system (1.1). Section 3 is devoted to the observer design. The proof of the main results are presented in Section 4. In Section 5, we apply the proposed EDO to deal with the high frequency disturbance. Numerical simulations are carried out in Section 7 to validate the theoretical results, followed up by conclusions in Section 8.

Throughout the paper, n and m denote the positive integers and $\mathbb{R}^n$ denotes the n-dimensional Euclidean space. The norm of $\mathbb{R}^n$ is denoted by $\|\cdot\|_{\mathbb{R}^n}$. The spectrum of operator or matrix A is denoted by $\sigma(A)$; the transpose of matrix A is represented by $A^\top$; $\Lambda_{\max}(A)$ stands for the maximal real part of eigenvalues of A. For simplicity, we denote $\mathbb{C}_+ = \{\lambda \in \mathbb{C} \mid \mathrm{Re}\lambda \geq 0\}$ and $\|\cdot\|_\infty = \|\cdot\|_{L^\infty[0,\infty)}$.

2 Observability

In this section, we consider observability for linear system with disturbance. Generally speaking, not all continuous disturbances can be estimated effectively in real time by a deterministic dynamic system. For instance, if the disturbance is a sample path of the Wiener process, it is differentiable for no time $t \geq 0$. In this case, we do not have any dynamic information about the disturbance and the estimation of such a disturbance via typical dynamic system observer seems impossible. Based on this observation, we first limit ourselves into an estimable signal space of the following:

$$W^{1,\infty}(\mathbb{R}^+) = \left\{ f : (0,+\infty) \to \mathbb{R} \text{ is locally integrable} \;\middle|\; f, \dot{f} \in L^\infty(0,+\infty) \right\}, \tag{2.1}$$

where the notation $\dot{f}(\cdot)$ can be regarded as the weak derivative of $f(\cdot)$ defined by

$$\int_0^{+\infty} \dot{f}(x)\phi(x)dx = -\int_0^{+\infty} f(x)\dot{\phi}(x)dx, \quad \forall\ \phi \in C_0^\infty(0,\infty), \tag{2.2}$$

where $C_0^\infty(0,\infty)$ stands for the space of infinitely differentiable functions with compact support in $(0,+\infty)$. By [1, p.249], $W^{1,\infty}(\mathbb{R}^+)$ is a Banach space equipped with the norm

$$\|f\|_{W^{1,\infty}(\mathbb{R}^+)} = \|f\|_\infty + \|\dot{f}\|_\infty, \quad \forall\ f \in W^{1,\infty}(\mathbb{R}^+). \tag{2.3}$$

Since the piecewise signal like

$$s_T(t) = \begin{cases} e^t, & t \in [0, T], \\ e^T, & t \geq T \end{cases} \tag{2.4}$$

belongs to $W^{1,\infty}(\mathbb{R}^+)$, the signal space $W^{1,\infty}(\mathbb{R}^+)$ is quite general and can include the harmonic signals, bounded continuously differentiable periodic signals, piecewise polynomial signals, piecewise exponential signals and their linear combinations.

Definition 1. Let $A \in \mathbb{R}^{n\times n}$, $B \in \mathbb{R}^n$ and $C \in \mathbb{R}^{1\times n}$. Suppose that Θ is a set of signals and we know that $d \in \Theta$. System (1.1) is said to be observable for the signal set Θ, provided both the initial state and the disturbance are distinguishable in the sense that: For any $T > 0$,

$$y(t) = 0 \ \text{ for a.e. } t \in [0, T] \ \Rightarrow \ x(0) = 0 \text{ and } d(t) = 0 \ \text{ for a.e. } t \in [0, T]. \tag{2.5}$$

From Definition 1, we see that system (1.1) is observable for Θ, it must have naturally that the (C, A) is an observable pair. However, Definition 1 is different from the observability of (C, A) for which the observability on some finite interval $[0, T]$ implies the observability on entire $[0, +\infty)$. Owing to the uncertainty of the disturbance, it is generally impossible to estimate the disturbance on $[T, +\infty)$ by the information of output over $[0, T]$.

Suppose that (C, A) is an observable pair. Then, system (1.1) is equivalent to its observability canonical form with

$$A = \begin{bmatrix} 0 & 0 & \cdots & 0 & a_1 \\ 1 & 0 & \cdots & 0 & a_2 \\ 0 & 1 & \cdots & 0 & a_3 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & a_n \end{bmatrix}, \quad B = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_{n-1} \\ b_n \end{bmatrix} \neq 0, \ C = [0 \ 0 \ \cdots \ 0 \ 1] \in \mathbb{R}^{1\times n}. \tag{2.6}$$

The following Lemma 2.1 characterizes the observability of system (1.1) for $W^{1,\infty}(\mathbb{R}^+)$.

Lemma 2.1. *Suppose that (C, A) is an observable pair. Then, system (1.1) is observable for $W^{1,\infty}(\mathbb{R}^+)$ if and only if its observability canonical form (2.6) satisfies $b_2 = b_3 = \cdots = b_n = 0$ and $b_1 \neq 0$.*

By Lemma 2.1, when system (1.1) is observable for $W^{1,\infty}(\mathbb{R}^+)$, system (1.1) with the observability canonical form (2.6) satisfies the familiar form

$$y^{(n)}(t) = a_1 y(t) + a_2 \dot{y}(t) + \cdots + a_n y^{(n-1)}(t) + b_1 d(t), \quad y(t) = x_n(t). \tag{2.7}$$

From now on, we always assume without loss the generality the following Assumption 2.1.

Assumption 2.1. *The pair (C, A) is observable and system (1.1) takes its observability canonical form (2.6) with $b_1 = 1$ and $b_i = 0, i = 2, 3, \cdots, n$.*

3 Extended dynamics observer

In this section, we shall design an observer for system (1.1) with $d \in W^{1,\infty}(\mathbb{R}^+)$. In many circumstances, $d(\cdot)$ is not completely unknown and may contain for instance some part that is generated by $(m+1)$-order differential equation of the following:

$$s^{(m+1)}(t) = g_0 s(t) + g_1 \dot{s}(t) + \cdots + g_m s^{(m)}(t), \quad g_i \in \mathbb{R},\ i = 0, 1, \cdots, m, \tag{3.1}$$

which can be written, like in the internal model principle, as an exosystem with the matrix G and its observation matrix E_1 as follows:

$$G = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ g_0 & g_1 & g_2 & \cdots & g_m \end{bmatrix} \in \mathbb{R}^{(m+1)\times(m+1)}, \quad E_1 = [1 \ \ 0 \ \ 0 \ \cdots \ 0]. \tag{3.2}$$

It is evident that the pair (E_1, G) is always observable. By [2], we define

$$\Omega(G) = \left\{ E_1 v(t) \middle| \dot{v}(t) = G v(t), v(0) \in \mathbb{R}^{m+1},\ t \geq 0 \right\} = \left\{ E_1 e^{Gt} v_0 \middle| v_0 \in \mathbb{R}^{m+1},\ t \geq 0 \right\}. \tag{3.3}$$

Then, $\Omega(G) \subset W^{1,\infty}(\mathbb{R}^+)$ as long as $\sigma(G) \subset i\mathbb{R}$ and each eigenvalue of G is algebraically simple.

Remark 3.1. It should be pointed that the properly chosen (E_1, G) is quite general for extracting the useful dynamics of the total disturbance $d(\cdot)$. Actually, let (F, S) be an arbitrary observable system with the state space $\mathbb{R}^{m+1}$ and output space $\mathbb{R}$. Define

$$\Omega(S) = \left\{ F v(t) |\ \dot{v}(t) = S v(t), v(0) \in \mathbb{R}^{m+1}, t \geq 0 \right\}. \tag{3.4}$$

Then all disturbance in $\Omega(S)$ with known S can be treated by the internal model principle. On the other hand, it follows from [2] that $\Omega(S)$ is independent of F and

$$\begin{aligned} \Omega(S) \quad &= \operatorname{span} \left\{ t^{n_\lambda - k} e^{\lambda t} \lambda \in \sigma(S), k = 1, \cdots, n_\lambda, t \geq 0, \right. \\ &\qquad \left. n_\lambda \text{ is the algebraic multiplicity of } \lambda \text{ as an eigenvalue of } S \right\}. \end{aligned} \tag{3.5}$$

As a result, $\Omega(S) = \Omega(G)$ provided that S is similar to G. When (F, S) is observable, there always exists an invertible matrix P such that $PSP^{-1} = G$ and $E_1 = FP^{-1}$. This implies that any useful dynamic information of the general total disturbance can be extracted by our special form (E_1, G) defined by (3.2).

Proposition 3.1. *Let (E_1, G) be given by (3.2) and $\Omega(G)$ be given by (3.3). Then, for any $Q \in \mathbb{R}^{1\times(m+1)}$ such that the pair (Q, G) is observable, it must have*

$$\Omega(G) = \left\{ Q v(t) \middle| \dot{v}(t) = G v(t), \forall\ v(0) \in \mathbb{R}^{m+1} \right\}. \tag{3.6}$$

For any $s \in W^{1,\infty}(\mathbb{R}^+)$, we denote by $\mathbb{P}_G s$ the projection of s on $\Omega(G)$, that is,

$$\mathbb{P}_G s = \arg \inf_{g \in \Omega(G)} \|s - g\|_{W^{1,\infty}(\mathbb{R}^+)}, \quad \forall\, s \in W^{1,\infty}(\mathbb{R}^+). \tag{3.7}$$

Since $\Omega(G)$ is a finite-dimensional subspace of the Banach space $W^{1,\infty}(\mathbb{R}^+)$, the $\mathbb{P}_G s$ must exist for any $s \in W^{1,\infty}(\mathbb{R}^+)$ ([22, p.35]).

In order to make use of the disturbance dynamics, we first need to write the disturbance dynamically.

Lemma 3.1. *Suppose that the matrix G defined by (3.2) satisfies $g_0 = 0$. Let (Q, G) be an observable pair with $Q \in \mathbb{R}^{1\times(m+1)}$. Then, for any $d \in W^{1,\infty}(\mathbb{R}^+)$, there exists a $v_0 \in \mathbb{R}^{m+1}$ such that*

$$\begin{cases} \dot{v}(t) = Gv(t) + \dfrac{B_d}{QB_d}\dot{e}(t), \quad v(0) = v_0, \\ d(t) = Qv(t), \end{cases} \tag{3.8}$$

where $B_d \in \mathbb{R}^{m+1}$ is the eigenvector corresponding to the eigenvalue 0 of G, $e = (I - \mathbb{P}_G)d$ and $\mathbb{P}_G$ is given by (3.7).

Given $d \in W^{1,\infty}(\mathbb{R}^+)$, we want to find a G by properly choosing $g_i, i = 0, 1, 2, \cdots, m$ in (3.2) such that the approximation error $\|(I - \mathbb{P}_G)d\|_{W^{1,\infty}(\mathbb{R}^+)}$ is as small as possible. The part $\mathbb{P}_G d \in \Omega(G)$ will be treated by the internal model principle through extended state observer, while the remaining part $\|(I - \mathbb{P}_G)d\|_{W^{1,\infty}(\mathbb{R}^+)}$ will be dealt with by the high-gain technique. Since the high-gain technique needs $0 \in \sigma(G)$, we assume that $g_0 = 0$ in Lemma 3.1. By Lemma 3.1, system (1.1) can be written dynamically as

$$\begin{cases} \dot{x}(t) = Ax(t) + BQv(t), \\ \dot{v}(t) = Gv(t) + \dfrac{B_d}{QB_d}\dot{e}(t), \\ y(t) = Cx(t). \end{cases} \tag{3.9}$$

where $e = (I - \mathbb{P}_G)d$ and $\mathbb{P}_G$ is given by (3.7). As a result, the observer design of system (1.1) comes down to the observer design of system (3.9). Hence, the EDO of system (1.1) with its canonical form (2.6) can be designed as

$$\begin{cases} \dot{\hat{x}}(t) = [A + (K_{\omega_o} + SE)C]\hat{x}(t) + BQ\hat{v}(t) - (K_{\omega_o} + SE)y(t), \\ \dot{\hat{v}}(t) = G\hat{v}(t) - EC\hat{x}(t) + Ey(t), \end{cases} \tag{3.10}$$

where ω_o denotes the positive tuning gain, $E = [0\ 0\ \cdots\ 1]^\top \in \mathbb{R}^{m+1}$, $K_{\omega_o} \in \mathbb{R}^n$ is a vector such that

$$\Lambda_{\max}(A + K_{\omega_o}C) = -\omega_o, \tag{3.11}$$

$S \in \mathbb{R}^{n\times(m+1)}$ and $Q \in \mathbb{R}^{1\times(m+1)}$ are the solutions of the equations

$$(A + K_{\omega_o}C)S - SG = BQ, \qquad CS = P_{\omega_o}, \tag{3.12}$$

with $P_{\omega_o} \in \mathbb{R}^{m+1}$ being a vector such that

$$\Lambda_{\max}(G + EP_{\omega_o}) = -\omega_o. \tag{3.13}$$

Let the observer errors be denoted by

$$\tilde{x}(t) = x(t) - \hat{x}(t) \quad \text{and} \quad \tilde{v}(t) = v(t) - \hat{v}(t). \tag{3.14}$$

Then, the errors are governed by

$$\begin{cases} \dot{\tilde{x}}(t) = [A + (K_{\omega_o} + SE)C]\tilde{x}(t) + BQ\tilde{v}(t), \\ \dot{\tilde{v}}(t) = G\tilde{v}(t) - EC\tilde{x}(t) + \dfrac{B_d}{QB_d}\dot{e}(t). \end{cases} \tag{3.15}$$

System (3.15) can be written as

$$\frac{d}{dt}(\tilde{x}(t), \tilde{v}(t))^\top = \mathcal{A}(\tilde{x}(t), \tilde{v}(t))^\top + \mathcal{B}\dot{e}(t), \tag{3.16}$$

where

$$\mathcal{A} = \begin{bmatrix} A + (K_{\omega_o} + SE)C & BQ \\ -EC & G \end{bmatrix}, \mathcal{B} = \frac{1}{QB_d}\begin{bmatrix} 0 \\ B_d \end{bmatrix}. \tag{3.17}$$

In terms of the solution S of the Sylvester equation of (3.12), we introduce the transformation

$$\mathbb{P} = \begin{bmatrix} I_n & S \\ 0 & I_{m+1} \end{bmatrix}, \tag{3.18}$$

where I_n and I_{m+1} are the identity matrices on $\mathbb{R}^n$ and $\mathbb{R}^{m+1}$, respectively. Thanks to the choice of K_{ω_o}, E and S, a simple computation shows that

$$\mathbb{P}\mathcal{A}\mathbb{P}^{-1} = \begin{bmatrix} A + K_{\omega_o}C & 0 \\ -EC & G + EP_{\omega_o} \end{bmatrix}, \tag{3.19}$$

which, together with (3.11) and (3.13), implies that the system matrix $\mathcal{A}$ of the error system (3.15) is Hurwitz. The equations (3.12) are very important in the observer design and their solvability is guaranteed by Lemma 4.1 in Section 4.

We are now in a position to summarize an implementable design scheme:

- Choose $K = [k_1 - a_1 \quad k_2 - a_2 \quad \cdots \quad k_n - a_n]^\top$ and $P = [p_0 \quad p_1 - g_1 \quad \cdots \quad p_m - g_m]$ such that the matrices $A + KC$ and $G + EP$ are Hurwitz. Let

$$K_{\omega_o} = [k_1\omega_o^n - a_1 \;\; k_2\omega_o^{n-1} - a_2 \;\; \cdots \;\; k_n\omega_o - a_n]^\top \tag{3.20}$$

  and

$$P_{\omega_o} = [p_0\omega_o^{m+1} \;\; p_1\omega_o^m - g_1 \;\; \cdots \;\; p_m\omega_o - g_m], \tag{3.21}$$

  where ω_o is a positive tuning parameter;

- Solve the equations (3.12) to obtain $S \in \mathbb{R}^{n\times(m+1)}$ and $Q \in \mathbb{R}^{1\times(m+1)}$.

Thanks to the choices of P and P_{ω_o}, a straightforward computation shows that

$$\lambda\omega_o \in \sigma(A + K_{\omega_o}C) \text{ if and only if } \lambda \in \sigma(A + KC). \tag{3.22}$$

Similarly, we also have

$$\lambda\omega_o \in \sigma(G + EP_{\omega_o}) \text{ if and only if } \lambda \in \sigma(G + EP). \tag{3.23}$$

By (3.22) and (3.23), we can change $\Lambda_{\max}(A + K_{\omega_o}C)$ and $\Lambda_{\max}(G + EP_{\omega_o})$ by adjusting the gain constant ω_o only. As a result, there is only one parameter that affects the performance of the observer. The mechanism of the observer design is based on the deep understanding about the internal model principle and the high-gain technique. Since the high-gain observer and the IMP-based observer have different structures. The combination of two different approaches is not straightforward.

Now, we state our main result Theorem 3.1.

Theorem 3.1. *In addition to the Assumption 2.1, suppose that the matrix G defined by (3.2) is diagonalizable, $g_0 = 0$ and*

$$\sigma(G) \subset \mathbb{C}_+. \tag{3.24}$$

Then, the EDO (3.10) for system (1.1) is well-posed: For any $d \in W^{1,\infty}(\mathbb{R}^+)$ and $(\hat{x}(0), \hat{v}(0)) \in \mathbb{R}^n \times \mathbb{R}^{m+1}$, there exists a positive constant M_1, independent of ω_o, such that

$$\lim_{t\to\infty} \|x(t) - \hat{x}(t)\|_{\mathbb{R}^n} \le \frac{M_1\|e\|_{W^{1,\infty}(\mathbb{R}^+)}}{\omega_o}, \tag{3.25}$$

where $e = (I - \mathbb{P}_G)d$ and $\mathbb{P}_G$ is given by (3.7). In particular, there exists a positive constant M_2, independent of ω_o, such that

$$\lim_{t\to\infty} |d(t) - Q\hat{v}(t)| \le \frac{M_2\|e\|_{W^{1,\infty}(\mathbb{R}^+)}}{\omega_o}. \tag{3.26}$$

Remark 3.2. We claim that the results in Theorem 3.1 can be actually applied to any system (1.1) that is observable for $W^{1,\infty}(\mathbb{R}^+)$. Actually, when system (1.1) is observable for $W^{1,\infty}(\mathbb{R}^+)$ but the Assumption 2.1 is not satisfied, by Lemma 2.1, one can always convert system (1.1) into the canonical form (2.6) for which the Assumption 2.1 is satisfied. Moreover, it follows from (3.8) and (4.51) below that the derivative of the approximation error $e(t)$ and its boundedness have been used in the proof of the Theorem 3.1. Hence, we should define in (3.7) the best approximation $\mathbb{P}_G$ with the norm $\|\cdot\|_{W^{1,\infty}(\mathbb{R}^+)}$ instead of the conventional Hilbert space norm $\|\cdot\|_{L^2(\mathbb{R}^+)}$ although the $L^2(\mathbb{R}^+)$ is complete and there exists a minimizer in dealing with the projection.

Remark 3.3. For any $d \in W^{1,\infty}(\mathbb{R}^+)$ and any matrix G, the disturbance can be divided into two parts:

$$d(t) = \mathbb{P}_G d + e \quad \text{with} \quad e = (I - \mathbb{P}_G)d, \tag{3.27}$$

where $\mathbb{P}_G d$ is given by (3.7). It is seen from observer (3.10) that it depends only on any given G satisfying (3.24) yet does not depend on the decomposition (3.27). The decomposition (3.27) is reflected only on the observation errors (3.25) and (3.26). We can improve the accuracy of the observer by proper choosing G and the choice of G depends on the prior information that we have known about the disturbance.

By (3.25), the accuracy of the observer depends both on the best approximation of $d(\cdot)$ on $\Omega(G)$ and the decay rate ω_o. This implies that there are two ways to improve the accuracy of the observer. First, choose the G so that $\Omega(G)$ is as large as possible to make the approximation error $\|e\|_{W^{1,\infty}(\mathbb{R}^+)}$ be as small as possible. Second, increase the gain ω_o. The choice of G depends on the prior information about the disturbance. The more accurate prior disturbance dynamics we have known, the smaller the steady-state error of the observer (3.10) will be. In particular, if we have known all the disturbance dynamics, i.e., we have known $d \in \Omega(G)$, the steady-state error becomes zero. Although the increasing of the gain ω_o can improve the observer accuracy, ω_o can not be, however, arbitrary large in engineering applications. This is the reason we need at all possible to make maximum use of known information of $d(\cdot)$ to reduce ω_0.

Corollary 3.1. *In addition to Assumption 2.1, suppose that the matrix G defined by (3.2) satisfies (3.24). Then, for any $d \in \Omega(G)$ and $(\hat{x}(0), \hat{v}(0)) \in \mathbb{R}^n \times \mathbb{R}^{m+1}$, the EDO (3.10) for system (1.1) satisfies*

$$\lim_{t\to\infty} e^{\beta t}\left[\|x(t)-\hat{x}(t)\|_{\mathbb{R}^n} + |d(t) - Q\hat{v}(t)|\right] = 0, \quad \forall\ 0 < \beta < \omega_o. \tag{3.28}$$

4 Proof of main results

In this section, we give proofs of the results stated in previous sections.

Proof of Lemma 2.1. Suppose that $b_1 \neq 0$ and $b_2 = b_3 = \cdots = b_n = 0$. Then, for any $T > 0$, $u(t) = 0$ for a.e. $t \in [0, T]$ implies that

$$\begin{cases} \dot{x}_1(t) = a_1 x_n(t) + b_1 d(t), \\ \dot{x}_2(t) = x_1(t) + a_2 x_n(t), \\ \cdots\cdots\cdots\cdots\cdots\cdots \\ \dot{x}_n(t) = x_{n-1}(t) + a_n x_n(t), \end{cases} \quad \text{a.e. } t \in [0, T], \tag{4.1}$$

where $x(t) = [x_1(t) \quad x_2(t) \quad \cdots \quad x_n(t)]^\top$. If $y(t) = x_n(t) = 0$ for a.e. $t \in [0, T]$, (4.1) yields $x_n(t) = x_{n-1}(t) = \cdots = x_1(t) = 0$, which implies that $d(t) = 0$ for a.e. $t \in [0, T]$ due to $b_1 \neq 0$. Hence, system (1.1) is observable for $W^{1,\infty}(\mathbb{R}^+)$.

Conversely, suppose that system (1.1) is observable for $W^{1,\infty}(\mathbb{R}^+)$. We first claim that $b_n = 0$.

Otherwise, for any $T > 0$,

$$\begin{cases} \dot{x}_1(t) = a_1 x_n(t) + b_1 d(t), \\ \dot{x}_2(t) = x_1(t) + a_2 x_n(t) + b_2 d(t), \\ \cdots\cdots\cdots\cdots\cdots\cdots \qquad\qquad \text{a.e. } t \in [0,T] \\ \dot{x}_n(t) = x_{n-1}(t) + a_n x_n(t) + b_n d(t), \\ y(t) = x_n(t) = 0, \end{cases} \tag{4.2}$$

implies that $x_{n-1}(t) + b_n d(t) = 0$ for a.e. $t \in [0,T]$ and hence system (4.2) turns out to be

$$\begin{cases} \dot{x}_1(t) = -\dfrac{b_1}{b_n} x_{n-1}(t), \\ \dot{x}_j(t) = x_{j-1}(t) - \dfrac{b_j}{b_n} x_{n-1}(t), \end{cases} \qquad j = 2, \cdots, n-1 \tag{4.3}$$

for a.e. $t \in [0,T]$. Since system (4.3) with the output $x_{n-1}(\cdot)$ is always observable for any $b_j \in \mathbb{R}$, $j = 1, 2, \cdots, n$, each non-zero solution of system (4.3) satisfies $x_{n-1}(t) = -b_n d(t) \neq 0$ for a.e. $t \in [0,T]$ and hence is the zero dynamics of the original system (4.2). This contradicts the observability of system (1.1) and hence $b_n = 0$. Similarly, we can obtain $b_{n-1} = \cdots = b_2 = 0$ by repeating the same process. This completes the proof of the lemma due to $B \neq 0$. ∎

Proof of Proposition 3.1. Since both (E_1, G) and (Q, G) are observable pairs, there exist invertible matrix P_1 and P_2 such that

$$\left(E_1 P_1^{-1}, P_1 G P_1^{-1}\right) = (Q_0, G_0) = \left(Q P_2^{-1}, P_2 G P_2^{-1}\right), \tag{4.4}$$

where (Q_0, G_0) is the observability canonical form. As a result,

$$E_1 = Q P_2^{-1} P_1, \quad G = P_1^{-1} P_2 G P_2^{-1} P_1, \tag{4.5}$$

and hence

$$E_1 e^{Gt} = Q P_2^{-1} P_1 e^{(P_1^{-1} P_2 G P_2^{-1} P_1)t} = Q P_2^{-1} P_1 P_1^{-1} P_2 e^{Gt} P_2^{-1} P_1 = Q e^{Gt} P_2^{-1} P_1. \tag{4.6}$$

This, together with (3.3), yields (3.6). ∎

Proof of Lemma 3.1. Since (Q, G) is an observable pair and $0 \in \sigma(G)$, it follows from the Hautus test [17, p.15, Remark 1.5.2] that $\mathrm{Ker} G \cap \mathrm{Ker} Q = \{0\}$. This means $QB_d \neq 0$ by the fact $GB_d = 0$ and $B_d \neq 0$. The first equation of (3.8) therefore makes sense. Since $\mathbb{P}_G d \in \Omega(G)$, it can be written dynamically as

$$\dot{v}_1(t) = G v_1(t), \quad (\mathbb{P}_G d)(t) = Q v_1(t) \tag{4.7}$$

for some initial state. Since $GB_d = 0$, by setting

$$v_2(t) = \frac{d(t) - (\mathbb{P}_G d)(t)}{QB_d} B_d = \frac{e(t)}{QB_d} B_d, \quad t \geq 0, \tag{4.8}$$

it has

$$\begin{cases} \dot{v}_2(t) = Gv_2(t) + \dfrac{B_d}{QB_d}\dot{e}(t), v_2(0) = \dfrac{e(0)}{QB_d}B_d, \\ e(t) = Qv_2(t). \end{cases} \tag{4.9}$$

Since $d \in W^{1,\infty}(\mathbb{R}^+)$, $e \in W^{1,\infty}(\mathbb{R}^+)$ as well. As a result, e is differentiable almost everywhere for $[0,\infty)$ [1, p.280,Theorem 5]. Hence, $\dot{e}$ in (4.9) makes sense. System (3.8) then follows from (4.7) and (4.9) by letting $v(t) = v_1(t) + v_2(t)$. ∎

Before prove Theorem 3.1, we first give a Lemma.

Lemma 4.1. *Under the assumptions of Theorem 3.1, the equations (3.12) are always solvable. Moreover, the following assertions are true:*

(i) The pair (Q, G) is observable;

(ii) For any $s \in \mathbb{C}_+$, there exist two positive constants C_K and C_A, independent of ω_o and s, such that

$$\|[s - (A + K_{\omega_o}C)]^{-1}B\|_{\mathbb{R}^n} \leq \frac{C_K}{\omega_o}, \tag{4.10}$$

$$\frac{1}{|QB_d|} = \frac{1}{|k_1 p_0|\omega_o^{n+m+1}} \tag{4.11}$$

and

$$\|C[s - (A + K_{\omega_o}C)]^{-1}\|_{\mathbb{R}^n} \leq \frac{C_A}{\omega_o^n}; \tag{4.12}$$

(iii) For any $s \in \mathbb{C}_+$, there exists a positive constant C_G, independent of ω_o and s, such that

$$\|[s - (G + EP_{\omega_o})]^{-1}E\|_{\mathbb{R}^{m+1}} \leq \frac{C_G}{\omega_o^{m+1}}, \quad \forall\, s \in \mathbb{C}_+; \tag{4.13}$$

(iv) If G is diagonalizable, then there exist two positive constants C_S and C_Q, independent of ω_o and s, such that

$$\|Sv\|_{\mathbb{R}^n} \leq C_S\|v\|_{\mathbb{R}^{m+1}}\omega_o^{m+n}, \quad \forall\, v \in \mathbb{R}^{m+1}, \tag{4.14}$$

and

$$|Q[s - (G + EP_{\omega_o})]^{-1}B_d| \leq C_Q\omega_o^{m+n}, \quad \forall\, s \in \mathbb{C}_+. \tag{4.15}$$

Proof. Since $A + KC$ is Hurwitz and by the choice of A, B, C and K_{ω_o}, a simple computation shows that $A + K_{\omega_o}C$ is Hurwitz as well. Since the matrix

$$A + K_{\omega_o}C = \begin{bmatrix} 0 & 0 & \cdots & 0 & k_1\omega_o^n \\ 1 & 0 & \cdots & 0 & k_2\omega_o^{n-1} \\ 0 & 1 & \cdots & 0 & k_3\omega_o^{n-2} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & k_n\omega_o \end{bmatrix} \tag{4.16}$$

is Hurwitz, (3.24) implies that

$$\sigma(A + K_{\omega_o}C) \cap \sigma(G) = \emptyset. \tag{4.17}$$

Moreover, a simple computation shows that

$$[\lambda - (A + K_{\omega_o}C)]^{-1}B = \frac{\mathcal{K}_\lambda}{\rho_A(\lambda, \omega_o)}, \quad \lambda \in \mathbb{C}_+, \tag{4.18}$$

where

$$\mathcal{K}_\lambda = \begin{bmatrix} k_n\lambda^{n-2}\omega_o + \cdots + k_3\lambda\omega_o^{n-2} + k_2\omega_o^{n-1} - \lambda^{n-1} \\ k_n\lambda^{n-3}\omega_o + \cdots + k_3\omega_o^{n-2} - \lambda^{n-2} \\ \vdots \\ k_n\omega_o - \lambda \\ -1 \end{bmatrix} \tag{4.19}$$

and

$$\rho_A(\lambda, \omega_o) = k_n\lambda^{n-1}\omega_o + \cdots + k_2\lambda\omega_o^{n-1} + k_1\omega_o^n - \lambda^n. \tag{4.20}$$

Hence, the following transmission zeros condition holds:

$$C[\lambda - (A + K_{\omega_o}C)]^{-1}B = \frac{-1}{\rho_A(\lambda, \omega_o)}, \quad \forall\ \lambda \in \mathbb{C}_+. \tag{4.21}$$

By [13], (4.17) and (4.21), the equations (3.12) are solvable.

(i). Suppose that $Gh = \lambda h$ and $Qh = 0$ with $\lambda \in \sigma(G)$. Then, the Sylvester equation in (3.12) turns out to be $(A + K_{\omega_o}C - \lambda)Sh = BQh = 0$. By (4.17), we conclude that $\lambda \notin \sigma(A + K_{\omega_o}C)$ and hence $A + K_{\omega_o}C - \lambda$ is invertible. As a result, $Sh = 0$ and $CSh = P_{\omega_o}h = 0$. By [17, p. Remark 1.5.2], we have $h = 0$ provided (G, P_{ω_o}) is an observable pair. Using [17, p. Remark 1.5.2] again, (Q, G) is an observable pair if we can prove so is for (G, P_{ω_o}). Actually, for any $Gv = \lambda v$ and $P_{\omega_o}v = 0$ with $\lambda \in \sigma(G)$, we have $(G + EP_{\omega_o})v = Gv = \lambda v$. Since $G + EP$ is Hurwitz, it follows from (3.2) and (3.21) that $G + EP_{\omega_o}$ is Hurwitz as well.

By (3.24) and the fact $\lambda \in \sigma(G)$, we obtain $\lambda \notin \sigma(G + EP_{\omega_o})$. Hence, $(G + EP_{\omega_o})v = \lambda v$ implies that $v = 0$. By [17, p. Remark 1.5.2], (G, P_{ω_o}) is an observable pair.

(ii). Since $A + K_{\omega_o}C$ is Hurwitz, it follows from (4.16) that $k_1 \neq 0$. The (4.10) can thus be obtained by (4.18), (4.19) and (4.20) easily. Since $GB_d = 0$, it follows from (3.12) that

$$(A + K_{\omega_o}C)SB_d = BQB_d \tag{4.22}$$

and hence

$$P_{\omega_o}B_d = CSB_d = C(A + K_{\omega_o}C)^{-1}BQB_d. \tag{4.23}$$

By (3.2) and (3.21),

$$P_{\omega_o}B_d = p_0\omega_o^{m+1}. \tag{4.24}$$

Since (Q, G) is an observable pair and $GB_d = 0$, the Hautus test [17, p.15, Remark 1.5.2] implies that $QB_d \neq 0$. We then combine (4.21), (4.24) and (4.23) to obtain $p_0\omega_o^{m+1} \neq 0$ and

$$\frac{1}{QB_d} = \frac{C(A + K_{\omega_o}C)^{-1}B}{P_{\omega_o}B_d} = \frac{1}{k_1p_0\omega_o^{n+m+1}}, \tag{4.25}$$

which leads to (4.11). In view of (4.16), a straightforward computation shows that, for any $s \in \mathbb{C}_+$,

$$C[s-(A+K_{\omega_o}C)]^{-1} = \frac{-1}{\rho_A(s,\omega_o)}[1 \quad s \quad \cdots \quad s^{n-1}],$$

which, together with (4.20), leads to (4.12).

(iii). By a straightforward computation, it follows that

$$G+EP_{\omega_o} = \begin{bmatrix} 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \\ p_0\omega_o^{m+1} & p_1\omega_o^m & \cdots & p_m\omega_o \end{bmatrix} \tag{4.26}$$

and hence

$$[s-(G+EP_{\omega_o})]^{-1}E = \frac{-1}{\rho_G(s,\omega_o)} \begin{bmatrix} 1 \\ s \\ \vdots \\ s^m \end{bmatrix}, \quad s \in \mathbb{C}_+, \tag{4.27}$$

where

$$\rho_G(s,\omega_o) = p_0\omega_o^{m+1} + p_1\omega_o^m s + \cdots + p_m\omega_o s^m - s^{m+1}. \tag{4.28}$$

Since $G+EP_{\omega_o}$ is Hurwitz, we have $p_0 \neq 0$ and so (4.13) follows from (4.27) and (4.28).

(iv). Since G is diagonalizable, for any $v \in \mathbb{R}^{m+1}$, there exists a sequence $v_0, v_1, v_2, \cdots, v_m$ such that $v = \sum_{j=0}^m v_j\varepsilon_j$, where $G\varepsilon_j = \lambda_j\varepsilon_j$ with $\lambda_j \in \sigma(G)$, $j = 0,1,2,\cdots,m$. By (3.12), (4.18) and (4.21), we have

$$S\varepsilon_j = (A+K_{\omega_o}C-\lambda_j)^{-1}BQ\varepsilon_j = -\frac{Q\varepsilon_j\mathcal{K}_{\lambda j}}{\rho_A(\lambda_j,\omega_o)} \tag{4.29}$$

and

$$\begin{aligned} P_{\omega_o}\varepsilon_j &= CS\varepsilon_j = C(A+K_{\omega_o}C-\lambda_j)^{-1}BQ\varepsilon_j \\ &= \frac{Q\varepsilon_j}{\rho_A(\lambda_j,\omega_o)}. \end{aligned} \tag{4.30}$$

Consequently,

$$Q\varepsilon_j = P_{\omega_o}\varepsilon_j\rho_A(\lambda_j,\omega_o), \quad j = 0,1,\cdots,m, \tag{4.31}$$

which, together with (4.29), gives

$$Sv = \sum_{j=0}^m v_jS\varepsilon_j = -\sum_{j=0}^m v_jP_{\omega_o}\varepsilon_j\mathcal{K}_{\lambda j}. \tag{4.32}$$

Combining (4.19), (3.21) and (4.32), we obtain (4.14).

Taking (4.26) and (3.2) into account, a simple computation shows that

$$[s-(G+EP_{\omega_o})]^{-1}B_d = \frac{1}{\rho_G(s,\omega_o)} \begin{bmatrix} p_ms^{m-1}\omega_o + p_{m-1}s^{m-2}\omega_o^2 + \cdots + p_1\omega_o^m - s^m \\ -p_0\omega_o^{m+1} \\ -p_0\omega_o^{m+1}s \\ \vdots \\ -p_0\omega_o^{m+1}s^{m-1} \end{bmatrix} \tag{4.33}$$

for any $s \in \mathbb{C}_+$. By (4.28) and the fact $p_0 \neq 0$, there exists a positive constant M_1, independent of ω_o and s, such that

$$\|G[s-(G+EP_{\omega_o})]^{-1}B_d\|_{\mathbb{R}^{m+1}} \leq M_1\|G\| \tag{4.34}$$

for any $s \in \mathbb{C}_+$. Consequently, it follows from (4.14), (4.12) and (4.34) that

$$|C[s-(A+K_{\omega_o}C)]^{-1}SG[s-(G+EP_{\omega_o})]^{-1}B_d| \leq C_S M_1\|G\|C_A\omega_o^m. \tag{4.35}$$

Combining (3.21), (4.28) and (4.33), there exists a positive constant M_2, independent of ω_o and s, such that

$$|P_{\omega_o}[s-(G+EP_{\omega_o})]^{-1}B_d| \leq M_2\omega_o^m, \quad \forall\ s \in \mathbb{C}_+. \tag{4.36}$$

For any $v \in \mathbb{R}^{m+1}$, it follows from (3.12) and (4.21) that

$$Qv = \frac{P_{\omega_o}v - C(A+K_{\omega_o}C)^{-1}SGv}{C(A+K_{\omega_o}C)^{-1}B} = -\rho_A(\lambda,\omega_o)[P_{\omega_o}v - C(A+K_{\omega_o}C)^{-1}SGv]. \tag{4.37}$$

As a result, there exists an M_3, independent of ω_o, such that

$$|Qv| = M_3\omega_o^n\Big[|P_{\omega_o}v| + |C(A+K_{\omega_o}C)^{-1}SGv|\Big], \quad \forall\ v \in \mathbb{R}^{m+1}, \tag{4.38}$$

which, together with (4.35) and (4.36), leads to (4.15). ∎

Proof of Theorem 3.1. Let the observer errors be (3.14). Then, they are governed by (3.15), or equivalently its abstract form (3.16). Let

$$\mathcal{A}_S = \begin{bmatrix} A+K_{\omega_o}C & 0 \\ -EC & G+EP_{\omega_o} \end{bmatrix}, \mathcal{B}_S = \begin{bmatrix} \dfrac{SB_d}{QB_d} \\ \dfrac{B_d}{QB_d} \end{bmatrix}. \tag{4.39}$$

A simple computation shows that

$$\mathbb{P}\mathcal{A}\mathbb{P}^{-1} = \mathcal{A}_S \ \text{ and } \ \mathcal{B}_S = \mathbb{P}\mathcal{B}, \tag{4.40}$$

where $\mathcal{A}, \mathcal{B}$ are those in (3.16), $\mathbb{P}$ is that in (3.18), and the Sylvester equation in (3.12) was used. For any $s \in \mathbb{C}_+$, a simple computation shows that

$$\mathbb{P}^{-1}(s-\mathcal{A}_S)^{-1}\mathcal{B}_S = \frac{1}{QB_d}\begin{bmatrix} [s-(A+K_{\omega_o}C)]^{-1}SB_d + SJ(s) \\ -J(s) \end{bmatrix}, \tag{4.41}$$

where

$$J(s) = -[s-(G+EP_{\omega_o})]^{-1}B_d + [s-(G+EP_{\omega_o})]^{-1}EC[s-(A+K_{\omega_o}C)]^{-1}SB_d. \tag{4.42}$$

Since $GB_d = 0$, it follows from (3.12) that

$$SB_d = (A+K_{\omega_o}C)^{-1}BQB_d, \tag{4.43}$$

and hence

$$\frac{[s-(A+K_{\omega_o}C)]^{-1}SB_d}{QB_d} = (A+K_{\omega_o}C)^{-1}[s-(A+K_{\omega_o}C)]^{-1}B. \tag{4.44}$$

By Lemma 4.1, there exist two positive constants C_K and C_S such that

$$\left\|\frac{[s-(A+K_{\omega_o}C)]^{-1}SB_d}{QB_d}\right\|_{\mathbb{R}^n} \leq \frac{C_K}{\omega_o^2}, \tag{4.45}$$

and

$$\frac{\|SJ(s)\|_{\mathbb{R}^n}}{|QB_d|} \leq \frac{C_S\|J(s)\|_{\mathbb{R}^{m+1}}\omega_o^{m+n}}{\omega_o^{n+m+1}} = \frac{C_S\|J(s)\|_{\mathbb{R}^{m+1}}}{\omega_o}, \quad \forall\ s\in\mathbb{C}_+. \tag{4.46}$$

By (4.28), (4.33) and the fact $p_0 \neq 0$, there exists a $C_J > 0$ such that

$$\|J(s)\|_{\mathbb{R}^{m+1}} < C_J, \quad \forall\ s\in\mathbb{C}_+. \tag{4.47}$$

Combining (4.46), (4.47), (4.42), (4.45), (4.41) and (4.11), we arrive at

$$\|\mathbb{P}^{-1}(s-\mathcal{A}_S)^{-1}\mathcal{B}_S\|_{\mathbb{R}^n} \leq \frac{C_{\mathcal{A}}}{\omega_o}, \quad \forall\ s\in\mathbb{C}_+, \tag{4.48}$$

where $C_{\mathcal{A}}$ is a positive constant independent of ω_o and s. Furthermore, it follows from (4.40) that

$$\|(s-\mathcal{A})^{-1}\mathcal{B}\|_{\mathbb{R}^n} = \|\mathbb{P}^{-1}(s-\mathcal{A}_S)^{-1}\mathcal{B}_S\|_{\mathbb{R}^n} \leq \frac{C_{\mathcal{A}}}{\omega_o} \tag{4.49}$$

for any $s\in\mathbb{C}_+$. Since both $A+K_{\omega_o}C$ and $G+EP_{\omega_o}$ are Hurwitz and satisfy (3.23) and (3.22), respectively, the matrix $\mathcal{A}$ is also Hurwitz. By (4.49), there exists an $L_{\mathcal{B}} > 0$, independent of ω_o, such that

$$\|e^{\mathcal{A}t}\mathcal{B}\|_{\mathbb{R}^n\times\mathbb{R}^{m+1}} \leq L_{\mathcal{B}}e^{-\omega_o t}, \quad t\geq 0. \tag{4.50}$$

We solve (3.16) to obtain

$$\begin{aligned}\|(\tilde{x}(t),\tilde{v}(t))\|_{\mathbb{R}^n\times\mathbb{R}^{m+1}} &= \left\|e^{\mathcal{A}t}(\tilde{x}(0),\tilde{v}(0))^\top + \int_0^t e^{\mathcal{A}(t-s)}\mathcal{B}\dot{e}(s)ds\right\|_{\mathbb{R}^n\times\mathbb{R}^{m+1}} \\ &\leq L_{\mathcal{A}}e^{-\omega_o t}X_0 + L_{\mathcal{B}}\int_0^t e^{-\omega_o(t-s)}\|\dot{e}\|_\infty ds \leq L_{\mathcal{A}}e^{-\omega_o t}X_0 + \frac{\|e\|_{W^{1,\infty}(\mathbb{R}^+)}L_{\mathcal{B}}}{\omega_o},\end{aligned} \tag{4.51}$$

where $X_0 = \|(\tilde{x}(0),\tilde{v}(0))\|_{\mathbb{R}^n\times\mathbb{R}^{m+1}}$ and $L_{\mathcal{A}}$ is a positive constant. This leads to (3.25) from (3.14).

Now, we prove (3.26). For any $s\in\mathbb{C}_+$, it follows from (4.41) that

$$\mathcal{Q}\mathbb{P}^{-1}(s-\mathcal{A}_S)^{-1}\mathcal{B}_S = -\frac{QJ(s)}{QB_d}, \quad \mathcal{Q}=(0,Q). \tag{4.52}$$

By (4.13), (4.12) and (4.14), there exists an $M_4 > 0$ such that

$$\|[s-(G+EP_{\omega_o})]^{-1}EC[s-(A+K_{\omega_o}C)]^{-1}SB_d\|_{\mathbb{R}^{m+1}} \leq \frac{M_4}{\omega_o}, \quad \forall\ s\in\mathbb{C}_+. \tag{4.53}$$

By (4.38), (3.21), (4.12) and (4.14), there exists an $M_5 > 0$ such that

$$|Qv| = M_5\omega_o^{n+m+1}\|v\|_{\mathbb{R}^{m+1}}, \quad \forall\ v\in\mathbb{R}^{m+1}. \tag{4.54}$$

We combine (4.53) and (4.54) to obtain

$$\left|Q[s-(G+EP_{\omega_o})]^{-1}EC[s-(A+K_{\omega_o}C)]^{-1}SB_d\right| \leq M_4M_5\omega_o^{n+m}, \quad \forall\ s\in\mathbb{C}_+,$$

which, together with (4.15), (4.42), (4.11) and (4.52), leads to

$$\left|\mathcal{Q}\mathbb{P}^{-1}(s-\mathcal{A}_S)^{-1}\mathcal{B}_S\right| = \left|\frac{QJ(s)}{|QB_d|}\right| \leq \frac{M_6}{\omega_o}, \tag{4.55}$$

where M_6 is a positive constant independent of ω_o and s. Owing to (4.40), we arrive at

$$|\mathcal{Q}(s-\mathcal{A})^{-1}\mathcal{B}| \leq \frac{M_6}{\omega_o}, \quad \forall\ s\in\mathbb{C}_+. \tag{4.56}$$

By (4.56) and the inverse Laplace transform, we obtain

$$\begin{aligned}\lim_{\omega_o\to+\infty}|\mathcal{Q}e^{\mathcal{A}t}\mathcal{B}| &= \frac{1}{2\pi i}\lim_{\omega_o\to+\infty}\lim_{T\to\infty}\int_{\gamma-iT}^{\gamma+iT}e^{st}\mathcal{Q}(s-\mathcal{A})^{-1}\mathcal{B}ds\\ &= \frac{1}{2\pi i}\lim_{T\to\infty}\int_{\gamma-iT}^{\gamma+iT}e^{st}\lim_{\omega_o\to+\infty}\mathcal{Q}(s-\mathcal{A})^{-1}\mathcal{B}ds\\ &= \frac{1}{2\pi i}\lim_{T\to\infty}\int_{\gamma-iT}^{\gamma+iT}e^{st}0ds = 0, \quad t>0,\end{aligned} \tag{4.57}$$

where γ is a real number so that the contour path of the integration is in the region of convergence of $\mathcal{Q}e^{\mathcal{A}t}\mathcal{B}$. Since the largest real part of eigenvalue of $\mathcal{A}$ is $-\omega_o$, (4.57) implies that

$$|\mathcal{Q}e^{\mathcal{A}t}\mathcal{B}| \leq L_Qe^{-\omega_o t}, \quad t\geq 0, \tag{4.58}$$

where L_Q is a positive constant which is independent of ω_o. As a result, the solution of system (3.15) satisfies

$$\begin{aligned}\lim_{t\to\infty}|Q\tilde{v}(t)| &= \lim_{t\to\infty}\left[\left|\mathcal{Q}e^{\mathcal{A}t}(\tilde{x}(0),\tilde{v}(0))^\top\right| + \left|\mathcal{Q}\int_0^t e^{\mathcal{A}s}\mathcal{B}\dot{e}(t-s)ds\right|\right]\\ &\leq \lim_{t\to\infty}\left|\int_0^t L_Qe^{-\omega_o s}|\dot{e}(t-s)|ds\right| \leq \frac{L_Q\|e\|_{W^{1,\infty}(\mathbb{R}^+)}}{\omega_o},\end{aligned}$$

which, together with (3.14) and (3.8), leads to (3.26). ∎

Proof of Corollary 3.1. Since $d\in\Omega(G)$, $e=(I-\mathbb{P}_G)d=0$. By Lemma 4.1, the pair (Q,G) is observable. It follows from Proposition 3.1 that $d(\cdot)$ can be represented dynamically to be governed by $\dot{v}(t)=Gv(t)$, $d(t)=Qv(t)$. Hence, the observer errors (3.14) are governed by

$$\frac{d}{dt}(\tilde{x}(t),\tilde{v}(t))^\top = \mathcal{A}(\tilde{x}(t),\tilde{v}(t))^\top, \tag{4.59}$$

where the matrix $\mathcal{A}$ is given by (3.17). Noting that both $A+K_{\omega_o}C$ and $G+EP_{\omega_o}$ are Hurwitz, $\mathcal{A}$ is Hurwitz as well due to the invertible transformation (3.19). ∎

5 Application to high frequency disturbance

The main advantage of the observer (3.10) lies in that we give a new way to improve the accuracy of the observer without increasing the high-gain ω_o. This is very useful when the disturbance contains high-frequency signals. If the disturbance $d(\cdot)$ contains the frequencies $\{\alpha_1, \cdots, \alpha_N\}$, we can choose $g_1, g_2, \cdots, g_{2N+1}$ such that the matrix G given by (3.2) with $m = 2N$ satisfies $\sigma(G) = \{0, \pm\alpha_j i \mid j = 1, 2, \cdots, N\}$. Thanks to the Vieta theorem, the choice of the parameters $g_0, g_1, g_2, \cdots, g_{2N+1}$ is easy and implementable. By (3.7), we have $\mathbb{P}_G d \in \Omega(G)$ and it follows from (3.25) that the steady-error of the observer (3.10) depends on the approximation error $e = (I - \mathbb{P}_G)d$ instead of the disturbance $d(\cdot)$ itself. Since the error $e(\cdot)$ never contains the frequencies $\{\alpha_1, \cdots, \alpha_N\}$ anymore, the negative impact of the signals with frequencies $\{\alpha_1, \cdots, \alpha_N\}$ has been completely eliminated by the extended dynamics determined by G.

Although we can design an observer for system (1.1) by the existing approaches such as the high-gain and the sliding mode methods, the performance of the observer may not be satisfactory provided the disturbance $d(\cdot)$ contains the high-frequency signals. This is because high-frequency parts of the disturbance usually leads to large $\|\dot{d}\|_\infty$ and the tuning gain of the observer can not be arbitrarily large in practice.

Next, we will give an example to show the main advantage of the proposed approach by comparing with the extended high-gain observer in [3]. For simplicity and without loss of the generality, we only consider system (1.1) with

$$A = \begin{bmatrix} 0 & a_1 \\ 1 & a_2 \end{bmatrix}, \ B = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \ \text{ and } \ C = [0 \ 1], \tag{5.1}$$

where $a_j \in \mathbb{R}$, $j = 1, 2$. Suppose that the disturbance contains the high-frequency α_0, i.e.,

$$d(t) = d_1(t) + d_0(t), \quad d_1 \in W^{1,\infty}[0, \infty), \quad d_0(t) = a_0 \sin \alpha_0 t + b_0 \cos \alpha_0 t, \tag{5.2}$$

where a_0 and b_0 are unknown amplitudes, α_0 may be very large. Suppose that all the prior knowledge about the disturbance we have known is that

$$\hat{\alpha}_0 \approx \alpha_0 \text{ where } \hat{\alpha}_0 \text{ is known.} \tag{5.3}$$

With the prior information (5.1) and (5.3), an extended dynamics observer of system (1.1) can be designed easily by (3.10) with

$$G = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -\hat{\alpha}_0^2 & 0 \end{bmatrix}. \tag{5.4}$$

By a simple computation, we have

$$\|d_0 - \mathbb{P}_G d_0\|_{W^{1,\infty}[0,\infty)} \le \|d_0 - (a_0 \sin \hat{\alpha}_0 t + b_0 \cos \hat{\alpha}_0 t)\|_{W^{1,\infty}[0,\infty)} \le M_0(\alpha_0 - \hat{\alpha}_0), \tag{5.5}$$

where $M_0 = \max\{\alpha_0, \hat{\alpha}_0\}(|a_0| + |b_0|)$. Consequently, the error $e(\cdot)$ of $d(\cdot)$ on $\Omega(G)$ satisfies

$$\|e\|_{W^{1,\infty}[0,\infty)} = \|d - \mathbb{P}_G d\|_{W^{1,\infty}[0,\infty)} \le \|d_1 - \mathbb{P}_G d_1\|_{W^{1,\infty}[0,\infty)} + M_0(\alpha_0 - \hat{\alpha}_0), \tag{5.6}$$

By (3.25) and (5.6), the upper boundedness of the steady-error of the EDO is

$$\frac{M_1\|d_1-\mathbb{P}_G d_1\|_{W^{1,\infty}[0,\infty)}+M_0(\alpha_0-\hat{\alpha}_0)}{\omega_o}, \tag{5.7}$$

where M_1 is a positive constant.

It is seen from (5.7) that the high-frequency signal $d_0(\cdot)$ has almost no effect on the steady-error of the EDO, provided the prior information is accurate enough, i.e., $|\alpha_0-\hat{\alpha}_0|$ is sufficiently small. In particular, when $\alpha_0=\hat{\alpha}_0$, the high-frequency signal $d_0(\cdot)$ does not affect the steady-error at all. Thanks to our extended dynamics G in (5.4), we can improve the accuracy of EDO without increasing the gain ω_o. However, the situation become completely different if we use the existing high-gain observer, see for instant the observer in [3]. Actually, the steady-error of the high-gain observer is $\frac{M_2\|\dot{d}\|_\infty}{\omega_o}$, where M_2 is a positive constant and ω_o is the tuning gain. Owing to the high-frequency content d_0 of the disturbance $d(\cdot)$, $\|\dot{d}\|_\infty$ can be very large. This implies that the tuning gain ω_o has to be much larger to ensure the accuracy of the high-gain observer. However, the large tuning gain ω_o may lead to poor transient response in practice.

6 Comparison with the extended state observer

If we choose specially $G=0$, then the disturbance dynamics can not be used in the observer (3.10). In this case, the EDO (3.10) will reduce to a high-gain observer automatically. For example, a simple computation will show that the observer (3.10) for system (1.1) with setting (5.1) and $G=0$ is

$$\begin{cases}\dot{\hat{x}}_1(t)=a_1\hat{x}_2(t)+\beta_1\beta_3\omega_o^3\hat{v}(t)-[(\beta_1-\beta_2\beta_3)\omega_o^2-a_1][y(t)-\hat{x}_2(t)],\\ \dot{\hat{x}}_2(t)=\hat{x}_1(t)+a_2\hat{x}_2(t)-[(\beta_2+\beta_3)\omega_o-a_2][y(t)-\hat{x}_2(t)],\\ \dot{\hat{v}}(t)=[y(t)-\hat{x}_2(t)],\end{cases} \tag{6.1}$$

where $\beta_3<0$ and the matrix $\begin{bmatrix}0 & \beta_1\\ 1 & \beta_2\end{bmatrix}$ is Hurwitz. In order to make a comparison to ESO [4] and the high-gain observer [3], we consider the canonical system

$$\begin{cases}\dot{z}(t)=A^\top z(t)+C^\top d(t),\\ y(t)=B^\top z(t),\end{cases}\qquad z(t)=[z_1(t)\quad z_2(t)]^\top, \tag{6.2}$$

where A,B and C are still given by (5.1). By virtue of the observer (6.1) and the invertible transformation

$$UAU^{-1}=A^\top,\ UB=C^\top,\ CU^{-1}=B^\top,\ \ U=\begin{bmatrix}0 & 1\\ 1 & a_2\end{bmatrix}, \tag{6.3}$$

the EDO of system (6.2) becomes

$$\begin{cases} \dot{\hat{z}}_1(t) = \hat{z}_2(t) - [(\beta_2+\beta_3)\omega_o - a_2][y(t) - \hat{z}_1(t)], \\ \dot{\hat{z}}_2(t) = a_1\hat{z}_1(t) + a_2\hat{z}_2(t) + \beta_1\beta_3\omega_o^3\hat{v}(t) \\ -[(\beta_1 - \beta_2\beta_3)\omega_o^2 + a_2(\beta_2+\beta_3)\omega_o - a_2^2 - a_1][y(t) - \hat{z}_1(t)], \\ \dot{\hat{v}}(t) = [y(t) - \hat{z}_1(t)]. \end{cases} \tag{6.4}$$

When $a_1 = a_2 = 0$, observer (6.4) is reduced to

$$\begin{cases} \dot{\hat{z}}_1(t) = \hat{z}_2(t) - (\beta_2+\beta_3)\omega_o[y(t) - \hat{z}_1(t)], \\ \dot{\hat{z}}_2(t) = \beta_1\beta_3\omega_o^3\hat{v}(t) - (\beta_1 - \beta_2\beta_3)\omega_o^2[y(t) - \hat{z}_1(t)], \\ \dot{\hat{v}}(t) = [y(t) - \hat{z}_1(t)], \end{cases} \tag{6.5}$$

and at the same time, system (6.2) turns out to be the canonical form of ESO in [4] or high-gain observer in [3]. In this case, the extended state observer or high-gain observer of system (6.2) is

$$\begin{cases} \dot{\hat{z}}_1(t) = \hat{z}_2(t) - \gamma_1\omega_o[y(t) - \hat{z}_1(t)], \\ \dot{\hat{z}}_2(t) = \ \hat{v}(t) - \gamma_2\omega_o^2[y(t) - \hat{z}_1(t)], \\ \dot{\hat{v}}(t) = -\gamma_3\omega_o^3[y(t) - \hat{z}_1(t)], \end{cases} \tag{6.6}$$

where γ_1, γ_2 and γ_3 are constants such that the following matrix:

$$\begin{bmatrix} -\gamma_1 & 1 & 0 \\ -\gamma_2 & 0 & 1 \\ -\gamma_3 & 0 & 0 \end{bmatrix}, \tag{6.7}$$

is Hurwitz. By proper choices of γ_1, γ_2 and γ_3, the observers (6.6) and (6.5) are equivalent under an invertible coordinate transformation. From this point, the proposed EDO with constant dynamic $G = 0$ covers the ESO as a special case and improves the ESO to the general observable linear system with input disturbance.

Remark 6.1. It follows from [3] that the conventional high-gain observer usually works for the chain of n integrators only. However, by Remark 3.2, the observer (3.10) works well for system (1.1) that is observable for $W^{1,\infty}(\mathbb{R}^+)$. In particular, the observer (3.10) with $G = 0$, as a new high-gain observer, also works well for system (1.1) that is observable for $W^{1,\infty}(\mathbb{R}^+)$. Hence, we actually propose a new high-gain observer that is is more general than the well-known high-gain observer in [3].

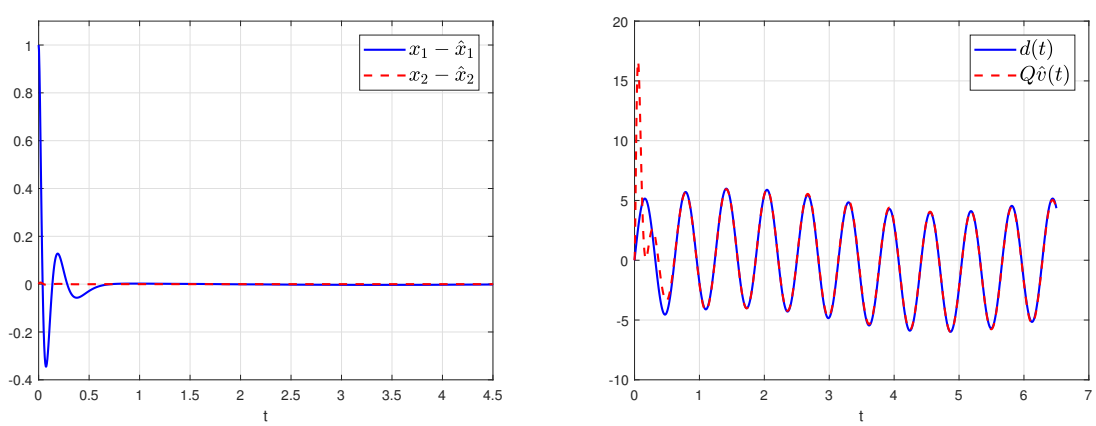
7 Numerical simulations

In order to validate the proposed principle visually, we present some simulations for the EDO (3.10). The finite difference scheme is adopted in discretization. The numerical results are programmed

in Matlab. The time step is taken as 0.001. Suppose that the plant is known and is given by (5.1) with $a_1 = 2$ and $a_2 = 1$. Let $(x_1(0), x_2(0)) = (0, 1)$ and let the initial state of the observer be zero. The disturbance is chosen as $d(t) = \sin t + \sin \alpha_0 t$ with $\alpha_0 = 10$. Suppose we have known that the disturbance contains the frequency $\alpha_0 = 10$. We must point out that "$\alpha_0 = 10$ is known" is just the known prior information rather than a required assumption. If this prior information is available, we can use it to improve the accuracy effectively. If it is unavailable, the EDO can still work. Comparing with the conventional ESO, we actually give an additional way to use the disturbance dynamics which can improve the observer accuracy. We will show this merit by numerical simulations in this section.

Since we have known the prior dynamics "$\alpha_0 = 10$", the extended dynamics matrix G of the observer (3.10) is chosen as (5.4) with $\hat{\alpha}_0 = \alpha_0 = 10$. The tuning gain is chosen as $\omega_o = 20$. Compared with the simulations in [3], the frequency of disturbance here is much larger but the tuning gain ω_o is relatively small. The error of EDO and the disturbance estimation are plotted in Figure 1. If we choose $G = 0$, EDO is reduced to the high-gain observer (6.1), which is equivalent to the conventional ESO by Section 6. The counterparts for the observer (6.1) with the same tuning gain $\omega_o = 20$ are plotted in Figure 2.

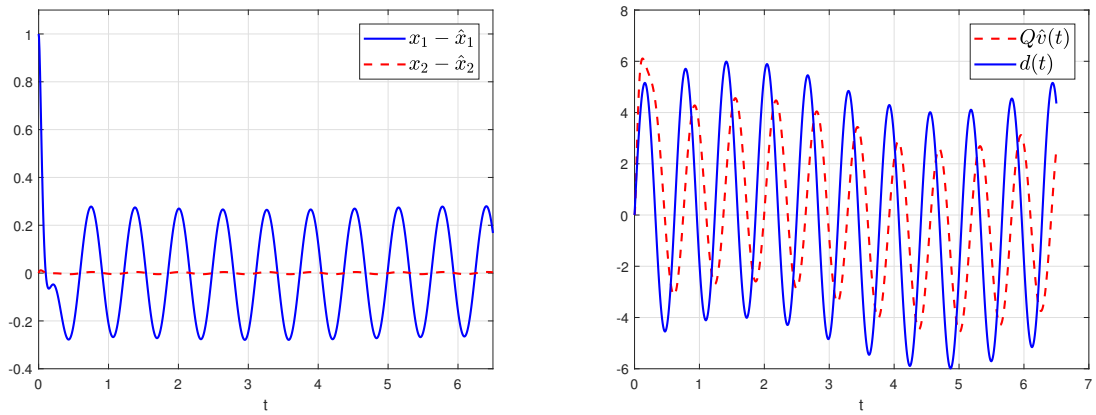
Since the observer gain ω_o is relatively small and the prior information "$\alpha_0 = 10$ is known" is wasted, the error of estimation is not satisfactory for the ESO (or equivalently, the high-gain observer). However, the accuracy of EDO can be improved significantly by using the prior information "$\alpha_0 = 10$". We increase the tuning gain of the ESO in Figure 3 by choosing $\omega_o = 100$. However, the steady-state error is still not satisfactory. Worse of all, a serious "peaking phenomenon" takes place during the transient time. Therefore, improving the accuracy of ESO by increasing the tuning gain only is not feasible. The EDO gives a way to reach the high observer accuracy with relatively small tuning gain ω_o by using the prior information "$\alpha_0 = 10$ is known". This implies that the EDO can tolerate more measuring noise than the ESO due to the small tuning gain ω_o.



(a) Error of state estimation

(b) $d(\cdot)$ and its estimation

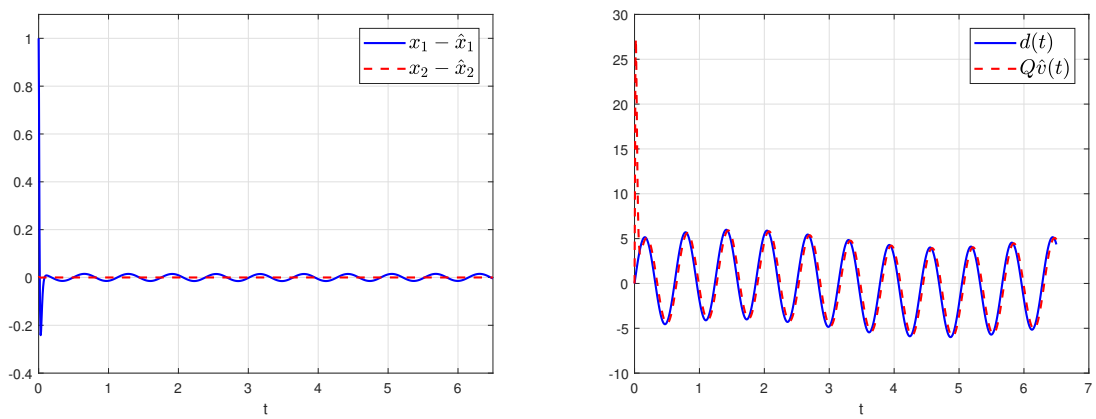
Figure 1: Estimation error of extended dynamics observer (3.10).



(a) Error of state estimation

(b) $d(\cdot)$ and its estimation

Figure 2: Estimation error of high-gain observer (6.1).



(a) Error of state estimation

(b) $d(\cdot)$ and its estimation

Figure 3: Estimation error of high-gain observer (6.1) with $\omega_o = 100$.

8 Conclusions

In this paper, a novel dynamics compensation approach is developed to estimate state of linear systems with disturbance. An extended dynamic observer (EDO) is designed, in terms of both the prior information and the online measurement information, to estimate both the disturbance and the system state simultaneously. The EDO takes almost all advantages from ESO and observer-based IMP. More specifically, it possesses strong robustness to the system and disturbance, as the ESO in ADRC, and at the same time, it proposes a feasible way to maximum utilize the prior information of the disturbance and the control plant. When there is no information about disturbance dynamics, the EDO is reduced automatically to an extension of ESO in ADRC which has been applied successfully to many engineering problems. Hence, the proposed EDO is quite general and can cover the high-gain observer and the IMP-based observer as special cases. Comparing with the

existing approaches such as ADRC and IMP, the proposed approach can improve the performance with relatively small tuning gain when the disturbance contains high-frequency signal. Therefore, the EDO makes up for the defects of the high-gain observer and the IMP-based observer. Since both of them have different structures, the design of EDO is our main contribution.

We just present a fundamental principle for the observer design. The technical tunings such as shaping the transient response are still required in engineering applications. From the theoretical point of view, this paper gives a systematic way to utilize the prior disturbance dynamics. The future works are the online computations of the disturbance dynamics by the measurements.

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